

CAMPUSCONNECT – INDIVIDUAL RESPONSIBILITY BREAKDOWN

SE2020 Module Deliverables & Contribution Documentation

SLIIT SE2020: Web and Mobile Technologies | Year 2 Semester 2 — 2026

PROJECT DELIVERABLES & MODULE BREAKDOWN

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Module: SE2020 — Web and Mobile Technologies
Degree: BSc (Hons) in Software Engineering (Year 2 Semester 2 — 2026)
Project Title: CampusConnect – Lost & Found Management System
GitHub Repo: <https://github.com/avishkasenanayake997-glitch/WMT.git>

1. ARCHITECTURE & BACKEND DEVELOPMENT

- Architected 3-tier system: React Native client, Express.js REST API, and MongoDB Atlas.
- Developed Mongoose schemas with strict validations for User, Item, and Claim entities.
- Created explicit relational foreign key references between Claim -> Item and User.
- Configured Multer diskStorage with MIME format filtering (JPEG/PNG/WebP) and 5MB size limits.
- Built centralized error handling middleware managing Mongoose CastError, duplicate keys, and 500s.
- Configured custom DNS resolver (Google 8.8.8.8) to bypass local router DNS blocking.

2. BUSINESS LOGIC & DATA CONSISTENCY

- Implemented Rule 1: Restricting claim submissions exclusively to Found items (HTTP 400).
- Implemented Rule 2: Enforcing Active item prerequisite before permitting claim creation (HTTP 400).
- Implemented Rule 3: Deduplicating pending claims per student/item (HTTP 409 Conflict).
- Implemented Rule 4 & 7: Atomic state updates on Admin approval (Claim -> Approved, Item -> Claimed).
- Implemented Rule 5: State isolation on Admin rejection (Item remains Active for other students).
- Implemented Rule 6: Claimant self-cancellation workflow (Pending -> Cancelled).

3. MOBILE APPLICATION & USER EXPERIENCE

- Built 14 functional React Native screens utilizing React Navigation (Stack + Bottom Tabs).
- Configured Expo SDK 57 for compatibility with iOS, Android, and Web browsers.
- Implemented AuthContext with persistent AsyncStorage for seamless JWT session restoration.
- Designed campus visual identity (Deep Blue #1E3A8A, Cyan #06B6D4) with status badges and cards.
- Built in-app administrative tools for claim review and item registry audit without web panels.
- Created one-tap demo credentials quick-fill on login screen for examiner viva evaluation.

4. VERIFICATION & DEPLOYMENT

- Authored comprehensive database seeder (node seeder.js) generating demo users and sample items.
- Programmed automated 10-point test suite (node test-api.js) verifying all 7 domain rules.
- Configured cloud deployment with render.yaml and MongoDB Atlas replica set connection.